PingSheng Kao

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SUMMARY

Applied AI Research Engineer and ML Data Scientist with 6+ years of experience prototyping LLM-enabled tools and evaluating multimodal, agentic, and production ML systems. Builds evals, human-calibrated judges, failure-analysis workflows, and technical feedback loops that turn ambiguous product questions into actionable model insights.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Built an internal LLM-assisted failure-analysis chat prototype combining natural-language investigation, VLM-as-judge retrieval of targeted vision-model failures, and Voxel51/FiftyOne image-collage visualization.
- Built simulated-user and agent-as-judge workflows to probe a multimodal conversational agent over long interactions, with targeted scenarios for visual-memory, sycophancy, factual-consistency, and tool-call failures.
- Developed a reference-based LLM evaluator and expert/non-expert validation workflow used internally for reporting and failure discovery; reached 90% expert alignment on a 40-example gold set.
- Translated user-research needs into engineering requirements and labeling instructions while coordinating UX research, engineering, Data Ops, experts, and human-in-the-loop validation.
- Built targeted agent scenarios whose systematic failure patterns changed subsequent development direction; scaled versioned LLM and agent judges on existing internal batch infrastructure and dashboards.
- Produced leadership-facing model-release reports translating deep-dive failure analysis and regression attribution into decision-ready metrics, including online validation of offline-mined patterns.

RESEARCH PUBLICATION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks ECCV 2026
Listed author (personal contribution: experiments) - Apple ML Research

- Listed author and experiment contributor on the ECCV 2026 publication GH-ESD, developing grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming & Data: Python, SQL, C++, JavaScript, AWS Athena
Applied AI: LLM/VLM Prototyping, Agent-as-Judge, LLM-as-Judge, Simulated Users, DSPy
Evaluation: Eval Design, Human Calibration, Tool-Call Evaluation, Scenario Generation, Failure Analysis
Methods & Tools: Embeddings, Clustering, OOD Analysis, Voxel51/FiftyOne, Tableau, Experiment Design